

Time Series Analysis: 304028011/354707001

Guideline for Final Data Analysis Report

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Purpose of the Report

The purpose of the final data analysis report is to demonstrate your ability to apply time series analysis methods to a real dataset. Your report should clearly describe the data, identify important time series patterns, fit appropriate models, assess model adequacy, and interpret the forecasting results in a practical context. The main focus should be on the problem, the data, the interpretation, and the practical implications, rather than simply presenting software output. **All figures, tables, and model results should be clearly explained.**

1 Title Page

Your title page should include the following information:

- Project title
- Student name
- Student ID
- Date of submission

2 Executive Summary

Provide a brief summary of the entire report. This section should be understandable to readers who may not be familiar with the technical details of time series analysis.

Your executive summary should include:

- The main research question
- The dataset used
- The major time series methods applied
- The key applied findings
- The final conclusion or recommendation

3 Introduction

3.1 Background

Describe the real-world context of your analysis. You may discuss:

- The phenomenon or business problem being studied
- Why the problem is important
- Who may use the results
- What decisions may be supported by the analysis

3.2 Objective of the Analysis

Clearly state the goal of your analysis in practical terms. The objective should address the needs of the target reader and explain how the analysis may support decision-making.

4 Data Description

4.1 Data Source

Describe where the data came from. You should include:

- Source of the data
- Time period covered
- Frequency of the data, such as daily, weekly, monthly, quarterly, or yearly
- Number of observations
- Description of the variables included

4.2 Data Cleaning

Explain any preprocessing steps performed before the analysis. Possible items include:

- Missing values
- Outliers
- Date formatting
- Transformations
- Other adjustments or corrections

When appropriate, briefly explain why these preprocessing steps were necessary.

5 Exploratory Data Analysis

5.1 Time Series Plot

Include a plot of the original time series. Discuss the following features:

- Overall pattern
- Increasing or decreasing trend
- Seasonal behavior
- Sudden changes
- Possible outliers
- Changes in variability over time

Interpret these features in relation to the application. For example, explain whether the observed patterns may reflect business cycles, seasonal demand, policy changes, unusual events, or other relevant real-world factors.

5.2 Summary Statistics

Report basic descriptive statistics, including:

- Mean
- Standard deviation
- Minimum
- First quartile
- Median
- Third quartile
- Maximum

Briefly interpret the results. Do not only report the numbers; explain what they suggest about the data. For example, comment on the central tendency, variability, skewness, and whether the minimum and maximum values are reasonable in the application context.

6 Model Identification

6.1 Stationarity Inspection

Discuss whether the series appears to be stationary. Consider:

- Whether the mean is stable over time
- Whether the variance is stable over time
- Whether a trend is present
- Whether seasonality is present

6.2 Transformation and Differencing

Explain any transformations used to make the series more suitable for modeling. Possible methods include:

- Box-Cox transformation
- First differencing
- Seasonal differencing
- Both regular and seasonal differencing

Briefly explain what the transformed or differenced series represents in practical terms. For example, first differencing may be interpreted as modeling changes from one period to the next.

6.3 ACF and PACF Plots

Include the autocorrelation function and partial autocorrelation function plots. Use these plots to discuss possible AR, MA, ARMA, ARIMA, or seasonal ARIMA structures.

6.4 Candidate Models

Describe the candidate models, including possible orders, considered in the analysis. Possible models include:

- AR model
- MA model
- ARMA model
- ARIMA model
- Seasonal ARIMA model

Briefly explain why each candidate model is considered.

6.5 Model Selection Criteria

State how the final model will be selected. Possible criteria include:

- AIC
- AICc
- BIC
- RMSE
- MAE
- MAPE

- Residual diagnostics
- Forecast performance

The final model should be selected based on both statistical evidence and practical usefulness.

7 Model Estimation

For each important model, report the estimated parameters. Briefly mention the estimation method used. Discuss whether the estimated parameters are meaningful and whether the fitted model is reasonable for the application.

8 Model Diagnostics

A good time series model should have residuals that behave approximately like white noise. Therefore, diagnostic checking is an essential part of the report.

8.1 Residual Plot

Include a time plot of the residuals. Discuss:

- Whether the residuals are centered around zero
- Whether there is any remaining trend
- Whether there is changing variance
- Whether there are outliers

8.2 ACF of Residuals

Include the ACF plot of the residuals. Discuss whether significant autocorrelation remains. If many residual autocorrelations are significant, the model may not be adequate.

8.3 Normality Check

If prediction intervals or statistical inference are discussed, check residual normality. Possible tools include:

- Histogram
- Q-Q plot

Briefly discuss whether the normality assumption appears reasonable.

9 Forecasting

State the forecast horizon and explain why this horizon is relevant to the application. Include a forecast plot with prediction intervals. Discuss:

- Forecasted trend
- Seasonal pattern
- Width of prediction intervals
- Practical interpretation of the forecast

The forecast should be interpreted in a way that connects directly to the original problem. Also comment on the uncertainty of the forecast, especially for longer forecast horizons.

10 Model Comparison

Compare the candidate models clearly.

Table 1: Model Comparison

Model	AICc	BIC	RMSE	MAE	...
ARIMA(1,1,1)					
SARIMA model					

Discuss which model is preferred and why. Do not select a model only because the software reports the smallest AIC or AICc. You should also consider whether the model produces reasonable residuals, accurate forecasts, and meaningful interpretation.

11 Conclusion

Summarize the main results. Your conclusion should include:

- The main patterns found in the data
- The final selected model
- The main forecasting results
- The practical implications
- Important limitations

The conclusion should directly answer the research objective stated in the introduction.

12 References

List all data sources, textbooks, papers, or websites used.

13 Appendix

Include supplementary materials if necessary. Possible items include:

- R or Python code
- Additional plots
- Additional model output
- Detailed calculations
- Data dictionary

Important Reminders for Students

Students should make sure that:

1. The report begins with a real-world problem, not with a model.
2. Every figure and table is explained in practical language.
3. The final model is justified using both statistical criteria and practical considerations.
4. Residual diagnostics are included.
5. Forecasts are interpreted together with uncertainty.
6. The report provides conclusions or recommendations relevant to the application.
7. The main text focuses on interpretation, while code and long software outputs should be placed in the appendix.